Presentation for companies

ISTENER Research Group

Universitat Jaume I

Research Team

Coordinator

Joaquín Navarro Esbrí – Full Prof.

Researchers (doctors)

Adrián Mota Babiloni – Assoc. Prof.

Ángel Barragán Cervera – Assoc. Prof.

Francisco Molés Ribera – External. Prof.

PhD Students

Pau Giménez Prades

Ghad Alarnaout Alarnaout

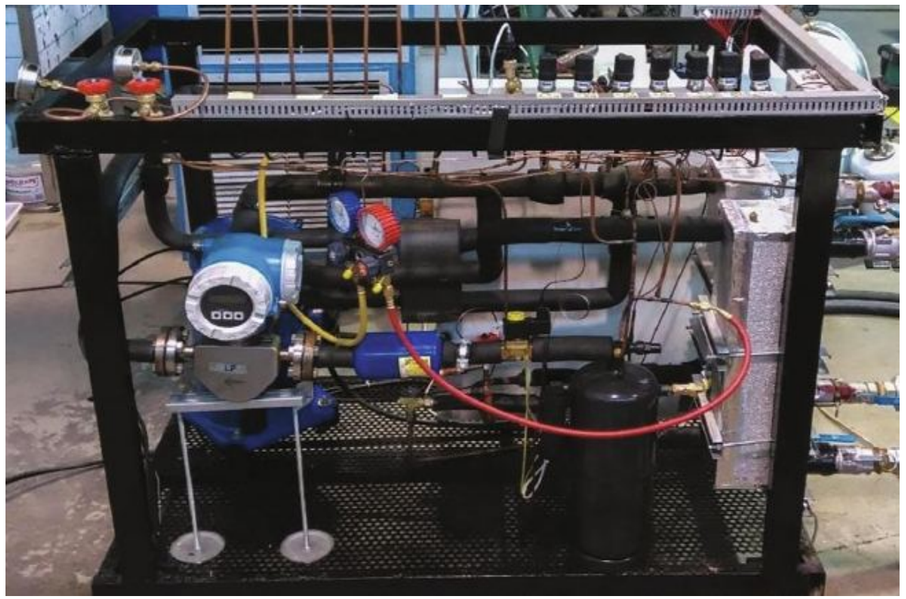
Alexandre Faus Redón

Alejandro Forés Palanques

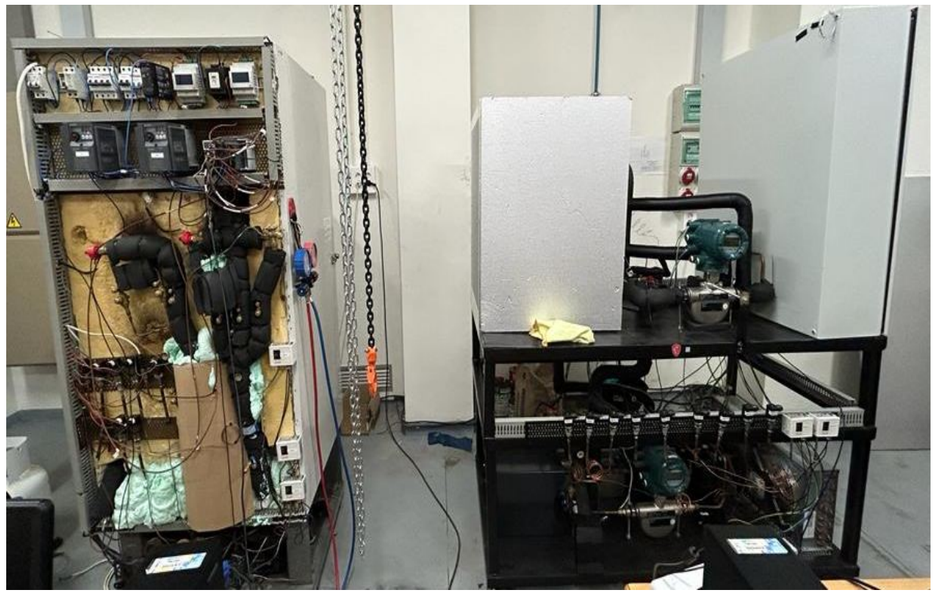


Main research lines

- Vapor compression systems (VCS) for cooling and refrigeration (ULT, LT, MT)
- Vapor compression systems for medium and high temperature heating (HTHP, MTHP, UHTHP)
- Waste heat valorization systems at low temperature (ORC)
- Environmentally friendly fluids/refrigerants for VCS and ORC







Refrigeration and air conditioning

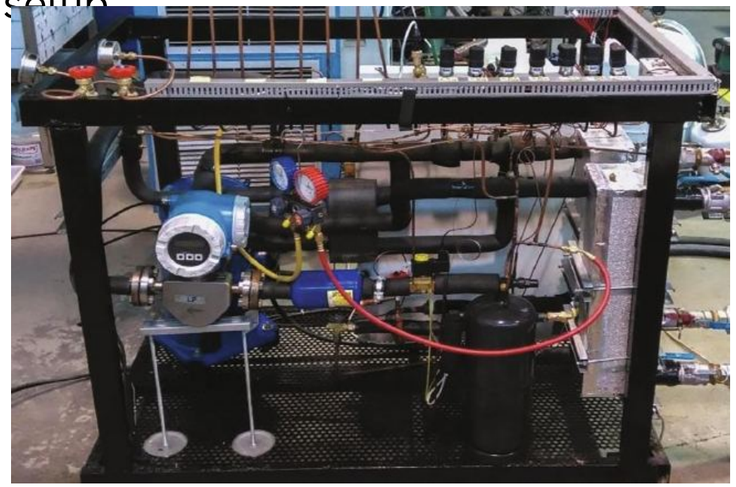
VAPOR COMPRESSION SYSTEMS FOR COOLING AND REFRIGERATION (ULT, LT, MT)

- Analysis of cooling generation systems from ULT refrigeration (-80ºC) to air conditioning applications
- Energy efficiency configurations and strategies
- Search and analysis of low GWP refrigerants

EXPERIMENTAL SETUPS

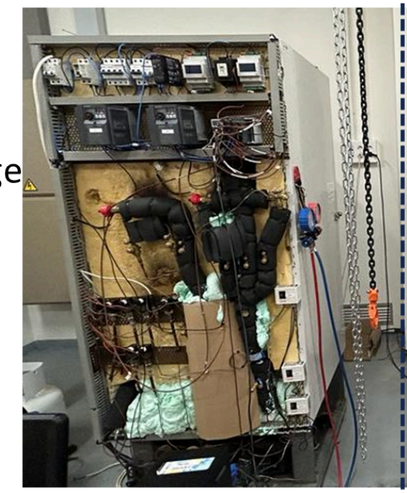
Up to 1500 W_c

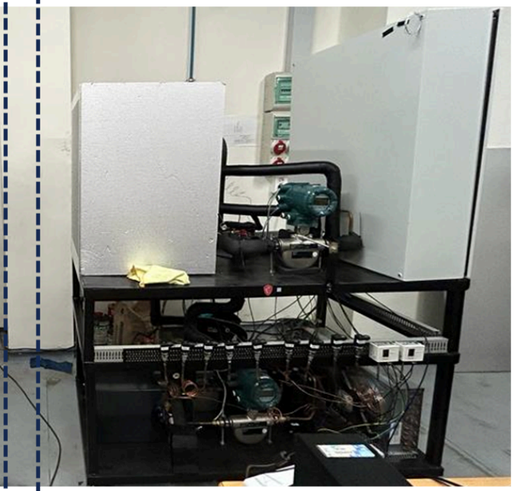
BT and MT regenerative single stage setup



5-15 kW_c

ULT two-stage cascade retrofit



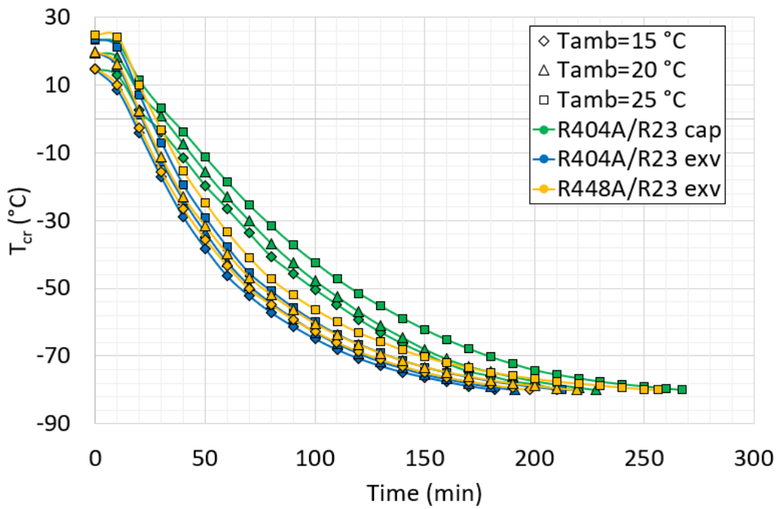


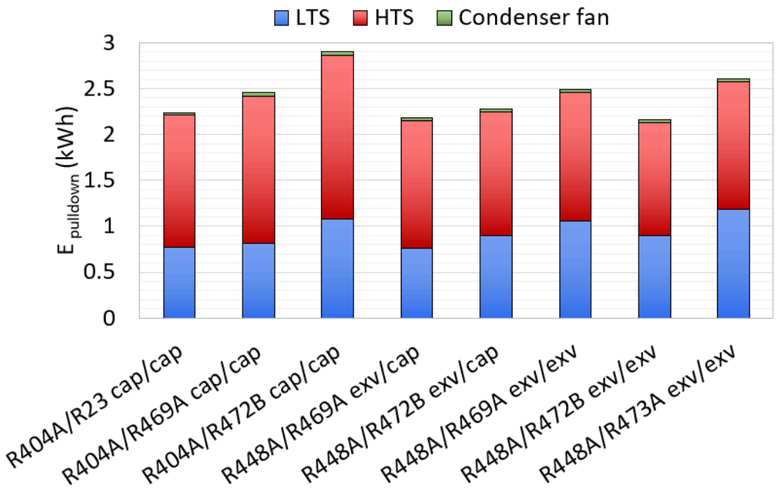
ULT two-stage cascade with IHX setup

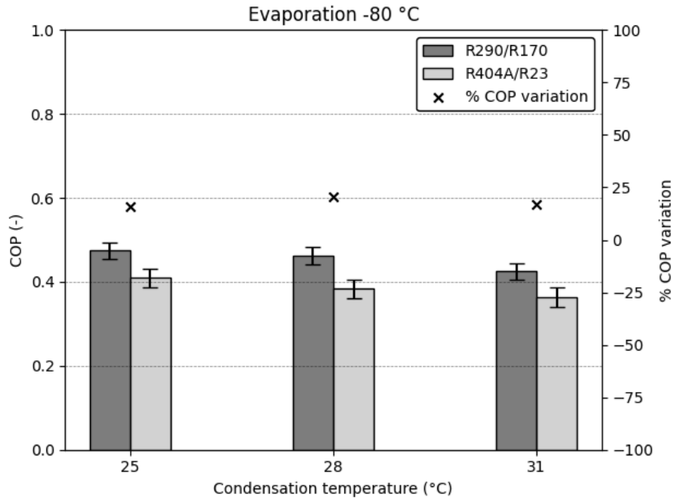
VAPOR COMPRESSION SYSTEMS FOR COOLING AND REFRIGERATION (ULT, LT, MT)

- Analysis of cooling generation systems from ULT refrigeration (-80ºC) to air conditioning applications
- Energy efficiency configurations and strategies
- Search and analysis of low GWP refrigerants

POSSIBLE ANALYSIS







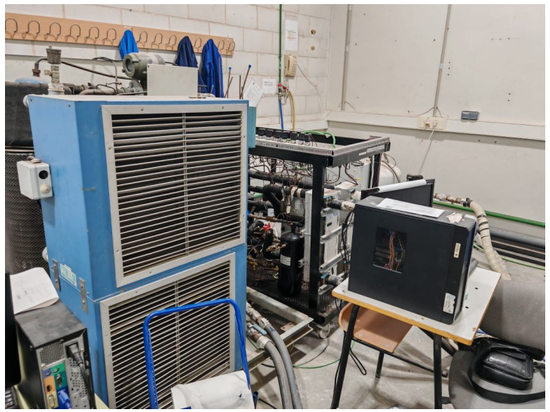
Heating production for medium and high temperature

VAPOR COMPRESSION SYSTEMS FOR MEDIUM AND HIGH TEMPERATURE HEATING (HTHP, MTHP, UHTHP)

- Analysis of heat generation systems from air conditioning to steam production (up to 180ºC)
- Energy efficiency configurations and strategies
- Search and analysis of low GWP refrigerants

EXPERIMENTAL SETUPS

Regenerative single stage setup HP



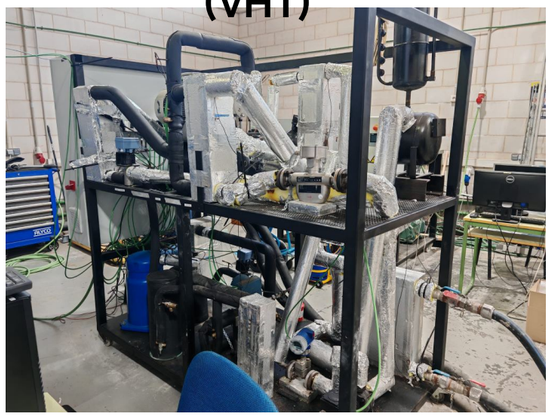
10-25 kW_H

Single stage regenerative setup HTHP (HT)



10-25 kW_H

Two-stage cascade regenerative setup HTHP (VHT)



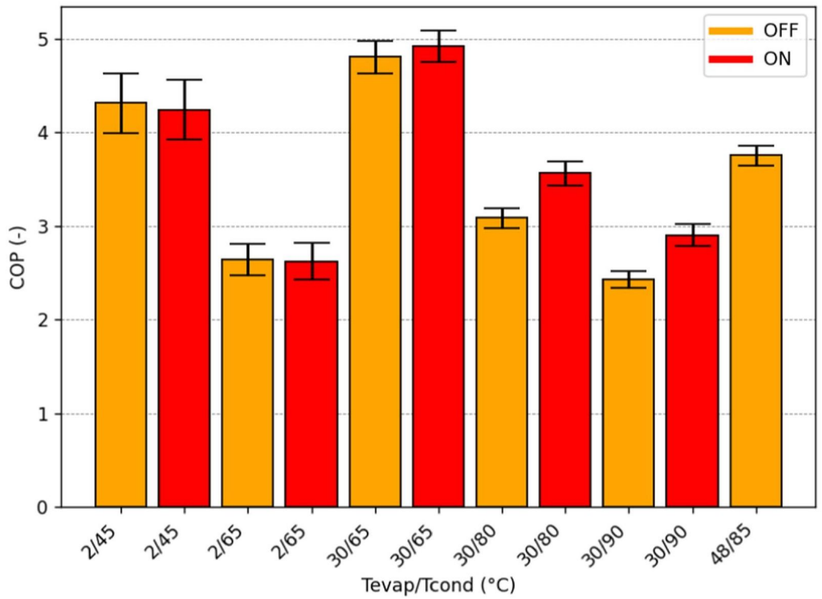
10-20 kW_H

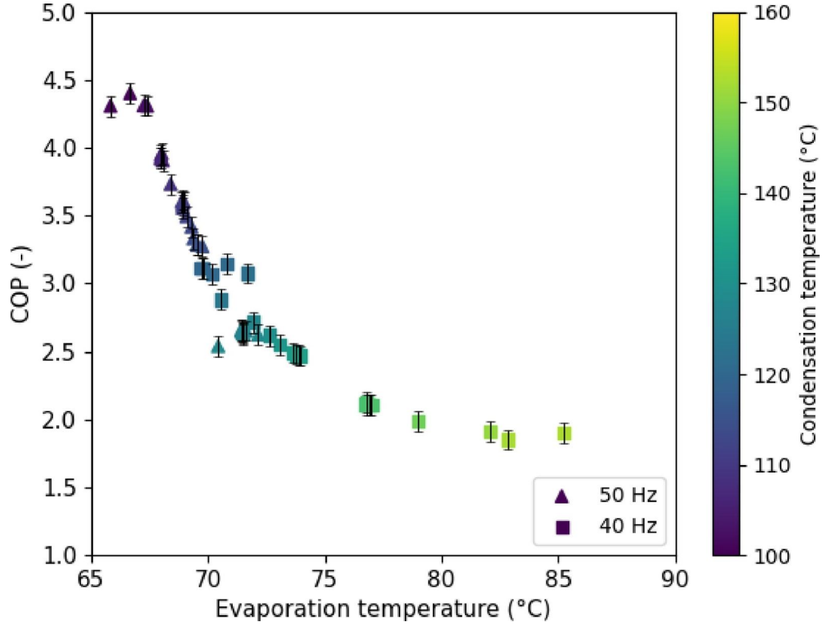
Heating production for medium and high temperature

VAPOR COMPRESSION SYSTEMS FOR MEDIUM AND HIGH TEMPERATURE HEATING (HTHP, MTHP, UHTHP)

- Analysis of heat generation systems from air conditioning to steam production (up to 180ºC)
- Energy efficiency configurations and strategies
- Search and analysis of low GWP refrigerants

POSSIBLE ANALYSIS





Waste heat recovery for electricity production

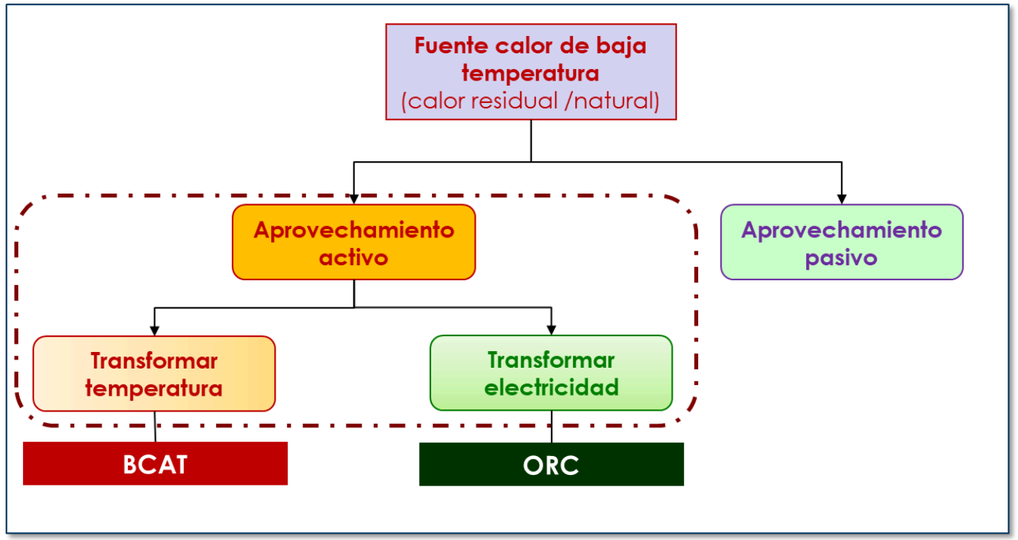
WASTE HEAT VALORIZATION SYSTEMS AT LOW TEMPERATURE (ORC)

- Analysis of electricity generation systems with low temperature heat sources (>80ºC)
- Energy efficiency configurations and strategies
- Search and analysis of low GWP refrigerants

EXPERIMENTAL SETUPS

Reversible HTHP-ORC setup HP



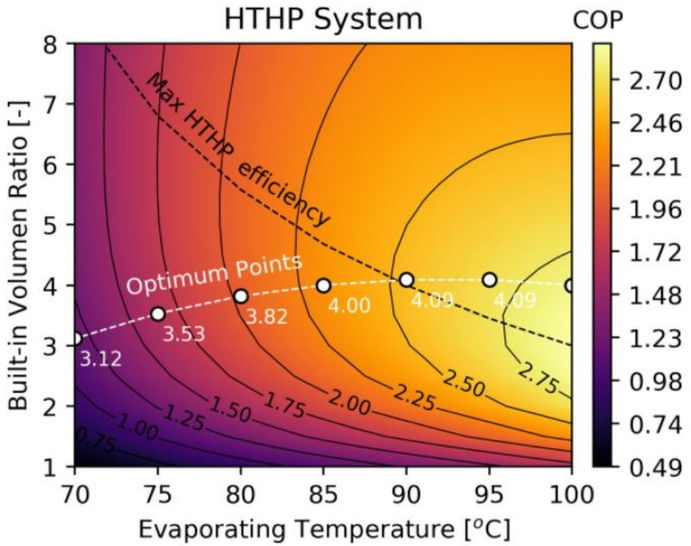


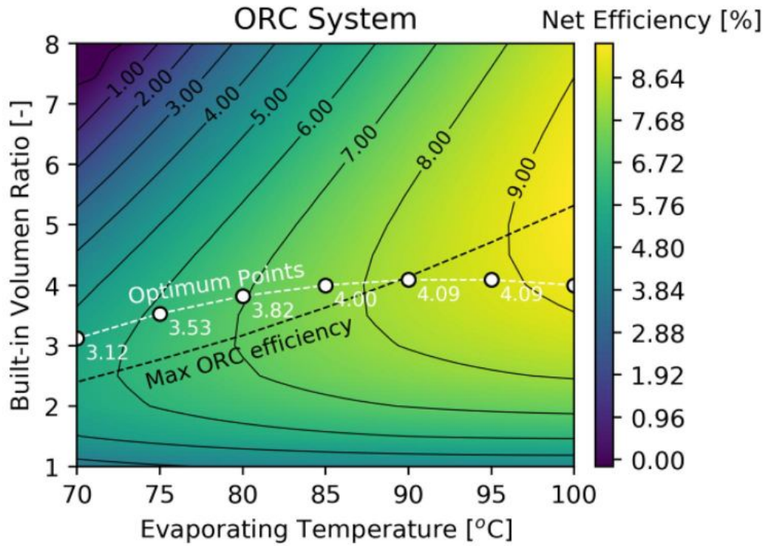
Waste heat recovery for electricity production

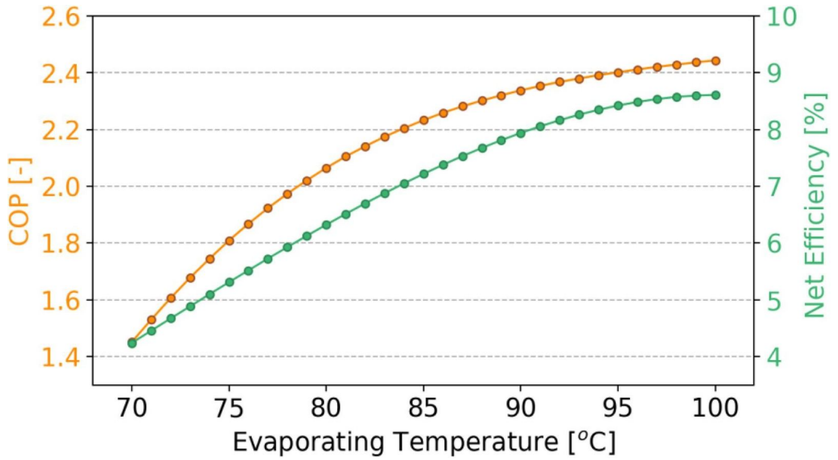
WASTE HEAT VALORIZATION SYSTEMS AT LOW TEMPERATURE (ORC)

- Analysis of electricity generation systems with low temperature heat sources (>80ºC)
- Energy efficiency configurations and strategies
- Search and analysis of low GWP refrigerants

POSSIBLE ANALYSIS



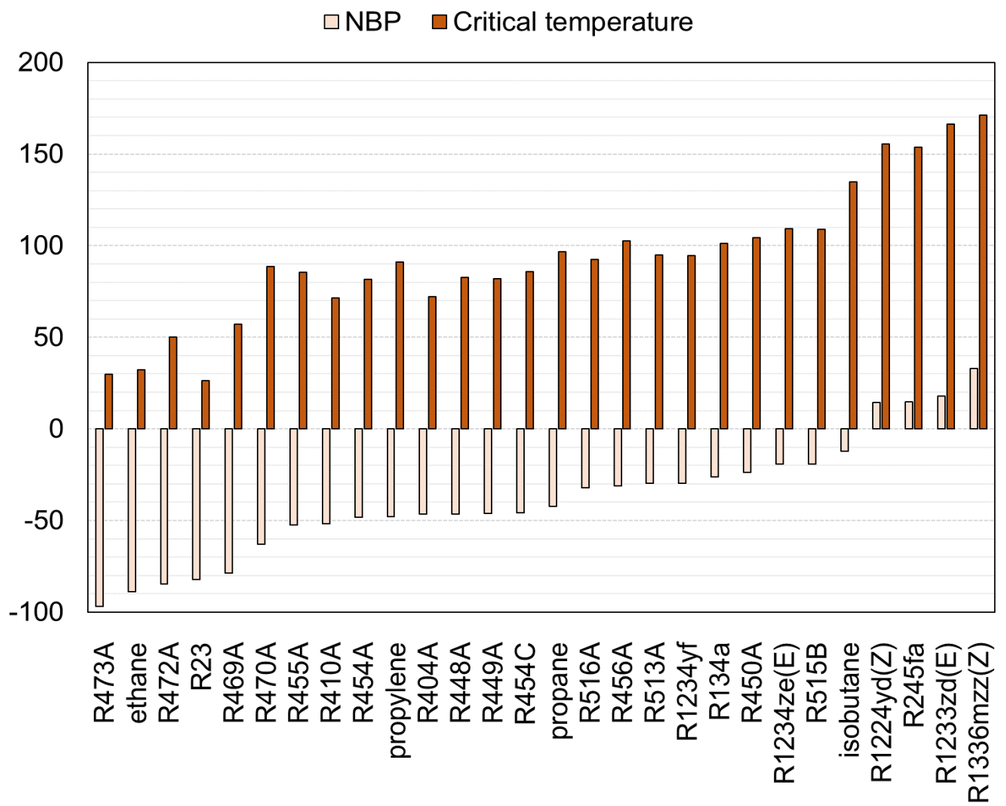




Future refrigerants

Environmentally friendly fluids/refrigerants for VCS and ORC

- Theoretical analysis of new fluids for the prediction of behavior in a real system.
- Experimental analysis of new fluids (new molecules and mixtures).
- Drop-in, retrofit and new design.
- Proposal of new mixtures for different applications.
- Subcritical/transcritical operation.
- Gas chromatography.



REFRIGERANTS WE HAVE TESTED

Laboratories and resources

HALF POWER TESTS WITH SECONDARY SYSTEMS

REFRIGERANT AND TOOL LAB



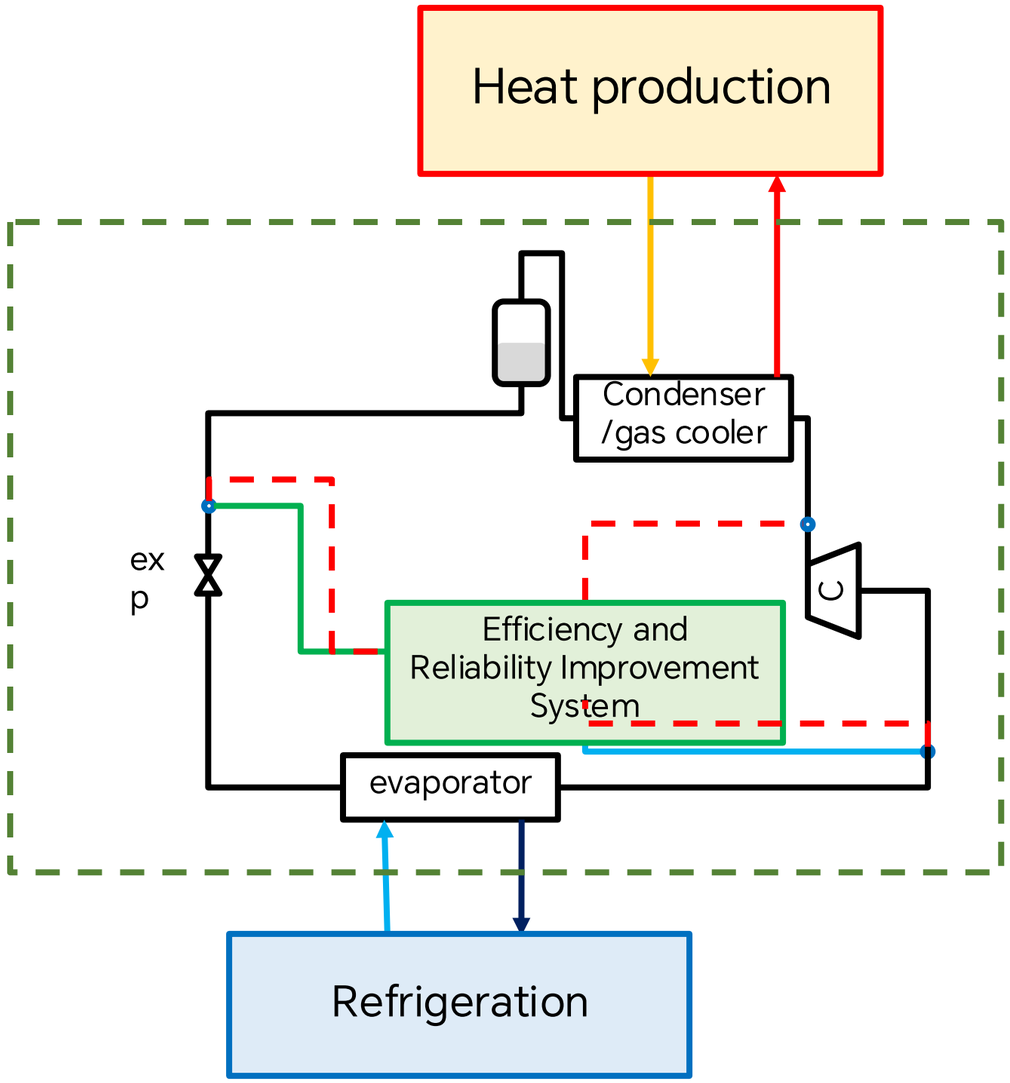
TESTING OF AUTONOMOUS EQUIPMENT WITH CONTROLLED TEMPERATURE

GAS CHROMATOGRAPH



Patente

EFFICIENCY AND RELIABILITY IMPROVEMENT SYSTEM



- Controlled increase in the cooling and/or heating effect (EER/COP)
- Control/limitation of critical component temperatures

It is based on a programmable physical system to maximize the required objectives, while maintaining control over critical component parameters.

Possible collaborations

We are open to signing an NDA for future discussions. We can adapt your format or ours

Refrigerants:
- Evaluation of new **natural and low-GWP refrigerants**.
- Study of **refrigerant blends** to optimize performance and safety.
- Analysis of **performance** and environmental impact.

Configurations
- Development of **dual and triple cascade** systems.
- Integration of **flash tanks, economizers, and intercoolers.**
- Optimization of **heat exchanger placement and control strategies.**

Products and components
- Collaboration in **product** development.
- Performance testing of **compressors and expansion devices**.
- **Heat exchanger design** for improved subcooling and stability.
- Assessment of **system integration and scalability** for different capacities.

CAPACITIES COULD BE EXTENDED WITH LONG-TERM COLLABORATION AND SUFFICIENT FUNDING



navarroj@uji.es
mota@uji.es

Presentation for companies

ISTENER Research Group

Universitat Jaume I